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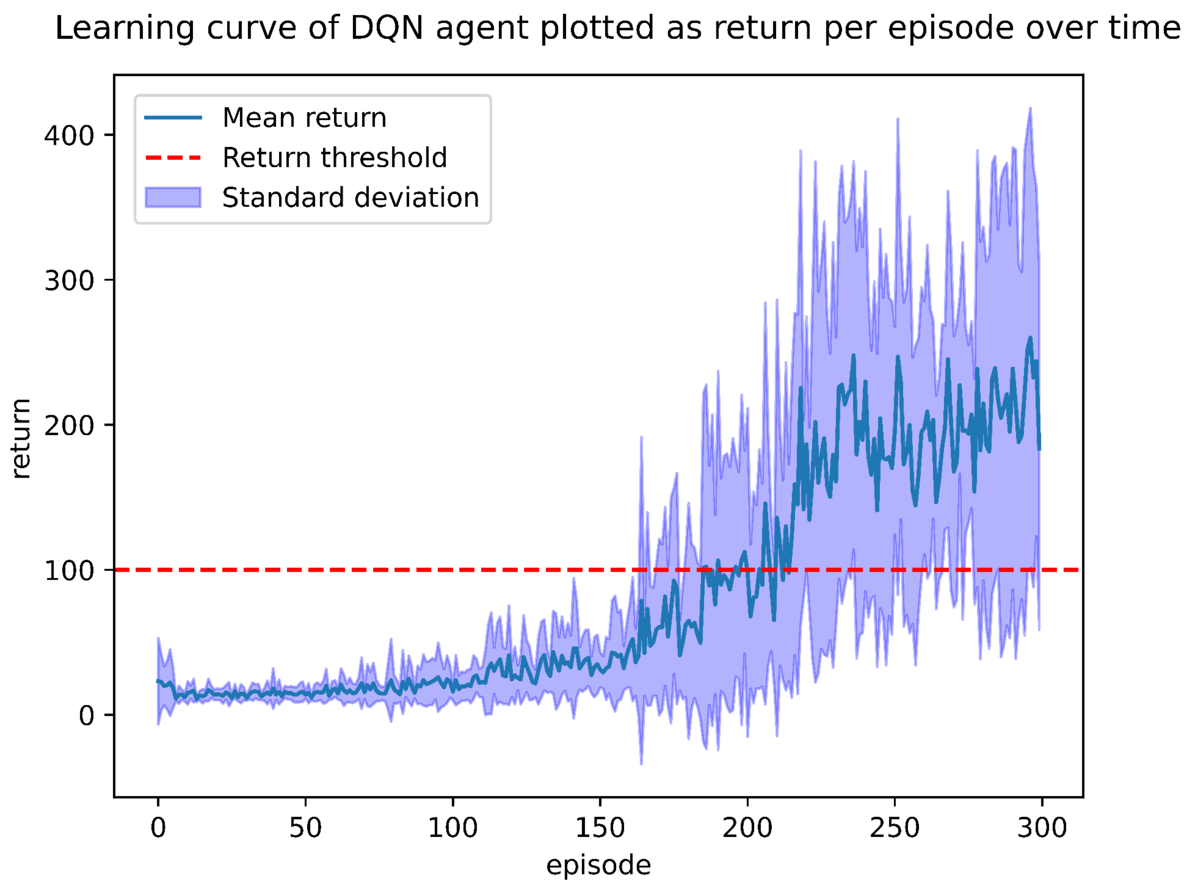
Question 1.1 - hyperparameters

Hyperparameter	**Value**	**Justification**
Architecture	1 hidden linear layer of size 50	Experiments show that a single layer of size 50 works better than one of size 25 or 100. Similarly, 2, 3 or 4 50-neuron layers perform worse.
Optimiser	Adam with default parameters (so learning rate = 10e-3)	Adam is a popular choice in literature due to its adaptive learning rate, fast convergence and few tunable parameters.
Epsilon	0.5 at the start, decays to 0.25, decay starts after 50 episodes	Works well in practice. Offers random exploration at the start. Leaves room for some randomness towards the end - this is needed as the cart force is random too.
Episodes	300	Works well in practice - high enough for convergence, low enough for low training times.
Replay buffer size	500	Large buffer size offers a smaller the probability of sampling correlated events hence the network is more stable.
Replay buffer batch size	8	Keeping the batch size a power of 2 makes it easy to parallelise the training over multiple cores.
Episodes between target network updates	25	A large value prevents run-away bias from bootstrapping and catastrophic forgetting

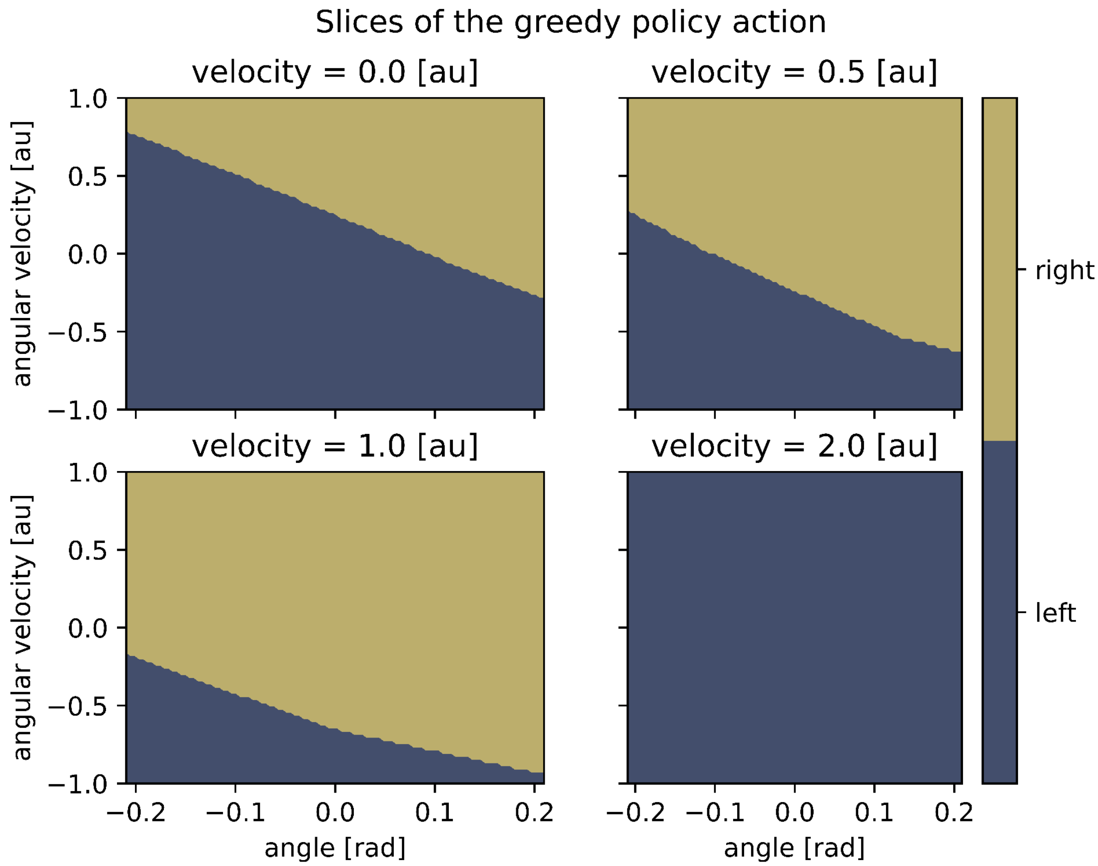
Question 1.2 - learning curve



Due to my choice of fixing epsilon = 0.5 for the first 50 episodes, the algorithm doesn't generate high returns in that period - this is a stage of pure exploration without exploitation. My reasoning behind this design choice was to minimise the chance of sampling correlated events and as a consequence minimise variance. For the next 100 episodes the returns remain roughly constant until they start growing rapidly around the 150^{th} episode. This relatively long convergence time can be attributed to the large 4-dimensional state space, the presence of 2 independent failure conditions (angle and distance) as well as the randomness in the force applied to the cart. As expected, the learning curve has a high standard deviation across training runs.

Note: the learning curve diagram visible in the Jupyter Notebook is different to the one presented above. The diagram presented in this report is the best result for my model while the diagram in the Jupyter Notebook was generated later due to chance it performs slightly worse.

Question 2.1 - slices of the greedy policy action

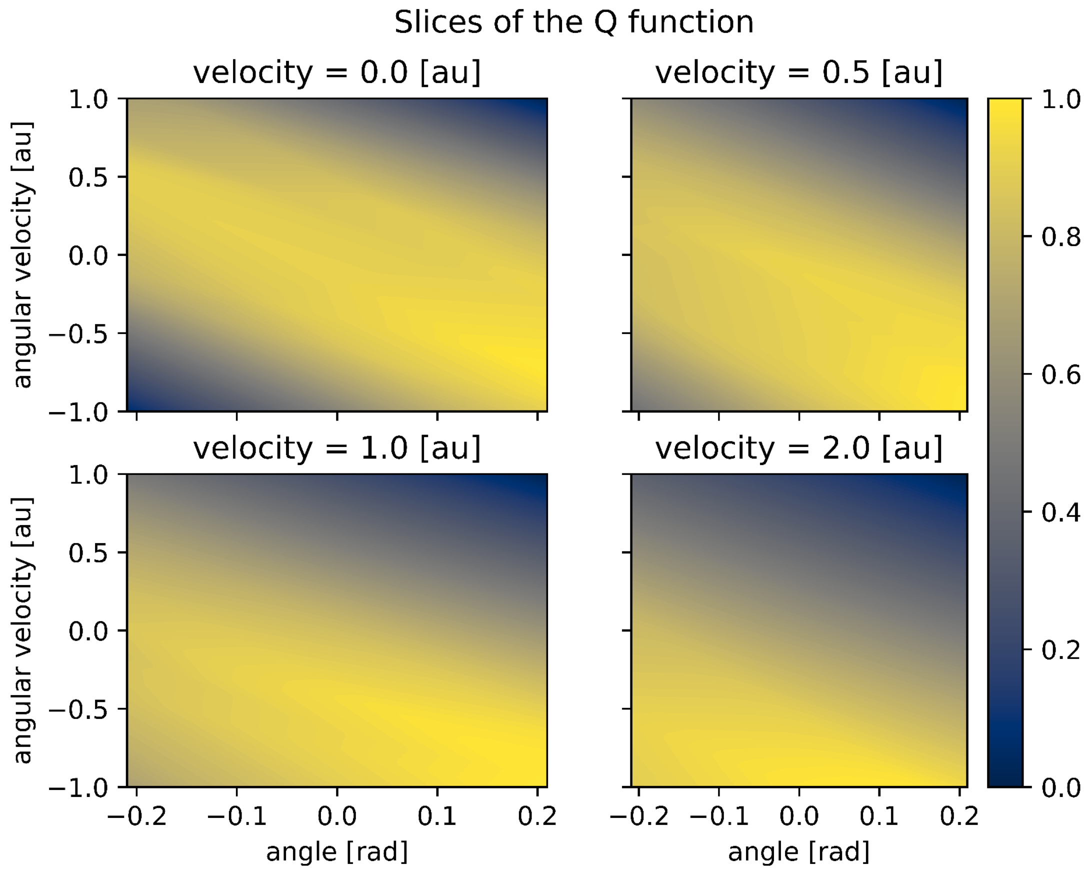


Cart position = 0.0 [au] for all above diagrams.

Before analysing the figures in this section (*slices of the greedy policy action* and *slices of the Q function*), let's observe the fact that both of them provide insight about the agent's strategy for keeping the pole upright rather than for maintaining its own position near the centre. The reason for this is that in both figures the cart position is fixed at the optimal value of 0.

The boundary between left and right policies is a downward slope. As either the angle or the angular velocity increases, the agent is more likely to go right. This makes sense because a positive angle means that the pole is leaning to the right and a positive angular velocity means the pole is actively falling to the right. Hence the cart needs to be moved to the right to keep the pole upright. Interestingly, as the velocity (in the right direction) increases, the agent increasingly prefers going right. At first glance this seems counterintuitive / wrong as it moves the cart away from the centre. However, examining the recorded video of the trained agent, it becomes clear that the agent has its own strategy for balancing the pole. When the pole is in its most unbalanced position, the agent wants to move the cart in a single swift move in the opposite direction - so the agent's goal is for the cart to reach velocity $\approx$ 1.5 as quickly as possible, and then when velocity $\approx$ 2.0, its goal is to abruptly change its direction because the pole has now become unbalanced in the other direction.

Question 2.2 - slices of the Q function



Cart position = 0.0 for all above diagrams.

Q-value represents the expected reward associated with being in a given state. So a cart in any of the bright yellow areas is likely to exist for a large number of timesteps, while the session of a cart in any of the dark blue areas is likely to be terminated soon. For an agent to have good chances of survival, when the angle increases, the angular velocity should decrease linearly. This makes sense as when the pole is tilted to one side, it's desirable that the pole be simultaneously be moving in the opposite direction to maintain balance. The cart is likely to be terminated if the angle and the angular velocity are either both positive or both negative. As the cart's velocity (in the right direction) increases, the optimal survival region shifts downwards, i.e. for any angle, it's now more desirable that the angular velocity be lower. This represents the fact that when the agent is making a successful maneouver to restore balance, it moves to the right and its pole moves to the left. On the other hand, when the agent is moving to the right and its pole is also moving to the right (angular velocity > 0), it is likely that the pole will soon fall.